A single forking token fixes the algorithm a code model writes

Abstract

Instruction-tuned code models write the same algorithm for a task almost every time, and sampling temperature is the knob usually turned to change that. We locate where in generation the algorithm is chosen, labelling programs by execution and resampling continuations from every prefix, and compare knobs under shared random seeds. A single forking token fixes the algorithm. On the three tasks on which Qwen3-8B already writes two algorithms, the algorithm's entropy given the class of that token was at most 0.189 bits (bootstrap 95% CI 0.123 to 0.239), down from 0.82 to 1.02 bits before it: a margin of −0.111 bits (95% CI −0.177 to −0.061) below the 0.3-bit bar of a preregistered test that all three tasks passed. A blind relocation found such tokens on two held-out gemma-3-4b-it tasks, but not on Llama-3.1-8B-Instruct's grid task. Temperature acts only through that token's odds: where the default leads there by several nats, no temperature from 0.3 to 1.3 changed any seed's algorithm, whereas random perturbations of the prompt's task specification states, which set those odds, changed it in 55% of seeds on two of those tasks, across one unperturbed and three perturbed runs (difference from temperature over seeds 0.545, 95% CI 0.41 to 0.67; preregistered). Predicting the size of the temperature effect from the logits of that token succeeded on only one of the three tasks. No knob, instruction, ban or activation edit produced a new algorithm: sampling knobs rephrase a solution, but the idea is chosen at one token among options the model already holds.

1 Introduction

Ask an instruction-tuned code model to implement a path search and it writes BFS, that is, breadth-first search; ask for a weighted shortest path and it writes Dijkstra's algorithm. Across six independent baselines, Qwen3-8B [Yang et al., 2025a] wrote BFS in 1,199 of 1,200 programs for grid path search and Dijkstra in 1,174 of 1,200 programs for shortest path [code] [code]. It wrote one algorithm in at least 49 of 50 programs on each of 13 further tasks derived from LeetCode [code], and eight more open-weight configurations wrote BFS or DFS, that is, depth-first search, on the grid task [code]. This is a code instance of generative monoculture [Wu et al., 2025a], also reported for other models [Narula et al., 2026, Lee et al., 2025]. It matters because the algorithm fixes a program's complexity and failure modes, and a model that always writes the textbook default cannot be asked for anything else by sampling it again.

The usual remedy is a knob: raise the sampling temperature [Peeperkorn et al., 2024], truncate the distribution differently [Nguyen et al., 2025], add randomness to the input [Nagarajan et al., 2025], or forbid the techniques of earlier solutions [Lu et al., 2025]. Whether such knobs change the idea or only its wording is rarely measured, because diversity metrics score text or rely on a language-model judge [Lee et al., 2025] and independent samples confound a knob with sampling noise. Code permits a sharper test: running two programs decides whether they implement the same algorithm, and reusing the same random numbers in every condition holds the noise fixed [Ravfogel et al., 2025].

Judging a knob also requires knowing where the choice is made. Resampling studies find forking tokens, positions at which resampling the continuation changes the outcome sharply [Bigelow et al., 2025], and reinforcement learning acts mainly at a few such high-entropy positions [Wang et al., 2025b, Meng et al., 2026], but these studies concern answers to reasoning problems. Where a code model chooses the algorithm it implements has not been located.

We locate that choice token by token, labelling every program by execution and comparing knobs on identical noise. The algorithm is decided at a single forking token: temperature can only reweight the alternatives that token already weighs, while the internal states of the prompt's task specification set its odds. Our contributions are the following.

- **One forking token fixes the algorithm (Section 5.1).** On the three mixed units, the Qwen3-8B tasks on which the model writes two algorithms, the algorithm's entropy given the class of one generated token is at most 0.189 bits (95% CI 0.123 to 0.239), a margin of -0.111 bits (95% CI -0.177 to -0.061) against a preregistered 0.3-bit bar that all three units passed. A blind relocation found such tokens on two held-out gemma-3-4b-it tasks, not on Llama-3.1-8B-Instruct's grid task.
- **Temperature only reweights that token (Section 5.2).** Where the default leads there by 4.85 to 7.46 nats, no temperature from 0.3 to 1.3 changed the algorithm of any of 20 seeds on three tasks, while phrasings multiplied; predicting the size of the temperature effect from the logits of that token worked on one of the three mixed units.
- **The task specification states set its odds (Section 5.3).** Random perturbations of those states changed the algorithm in 55% of seeds on two tasks, across one unperturbed and three perturbed runs (pooled difference from temperature over seeds 0.545, 95% CI 0.41 to 0.67; preregistered), and patching in the states of a prompt that names the alternative wrote it on 30 of 30 seeds.
- **Post-training widened the margin at that token (Section 5.4)** by 2.11 to 13.41 nats over Qwen3-8B-Base on three collapsed units, on which the model writes one algorithm.
- **No route produced a confirmed new algorithm (Sections 5.5 and 5.6).** Context seeds, novelty instructions and bans moved the choice only among known algorithms, and none of 45,892 audited correct programs was new; a program search with an evaluator gave eight programs one reader classed as new, which its replication did not reproduce.

Figure 1 traces the account on minimum spanning tree weight, with shortest path for the temperature margin, and Figure 2 shows the measurement it rests on.

2 Related Work

Algorithmic diversity of code models. Generative monoculture names the narrowing of output diversity relative to what a task admits, code included [Wu et al., 2025a]. Lee et al. [2025] cluster correct solutions by algorithm with a language-model judge and find low algorithmic diversity that rises with temperature above 1 at a cost in correctness; prompt cues steer the choice [Narula et al., 2026], denial prompting forbids earlier techniques [Lu et al., 2025], and implementation priors explain why constraints against a default fail [Tang, 2026]. We label the mechanism a program executes and locate where the choice is made.

Creativity knobs. Temperature correlates only weakly with novelty in stories [Peeperkorn et al., 2024]. Min-p sampling [Nguyen et al., 2025], dynamic temperature driven by entropy [Zhang et al., 2024] and adaptive temperature for code [Zhu et al., 2024] reshape the next-token distribution; seed-conditioning [Nagarajan et al., 2025], intent factored generation [Ahmed et al., 2025] and PlanSearch [Wang et al., 2025a] add randomness or plans before the answer; chat templates themselves collapse diversity [Yun et al., 2025]. We compare knobs on shared sampling noise, as Gumbel counterfactual generation does [Ravfogel et al., 2025].

Forking tokens and post-training. Forking Paths finds tokens at which the outcome distribution of resampled continuations changes abruptly [Bigelow et al., 2025]; hidden states anticipate such outcomes [Zur et al., 2025], and high-entropy minority tokens [Wang et al., 2025b] and thought anchors [Bogdan et al., 2025] carry the decisions of reasoning traces. Reinforcement learning mostly reweights choices the base model already considers [Meng et al., 2026], most tokens of an aligned model are the base model's top token [Lin et al., 2024], alignment can sit in a response's first

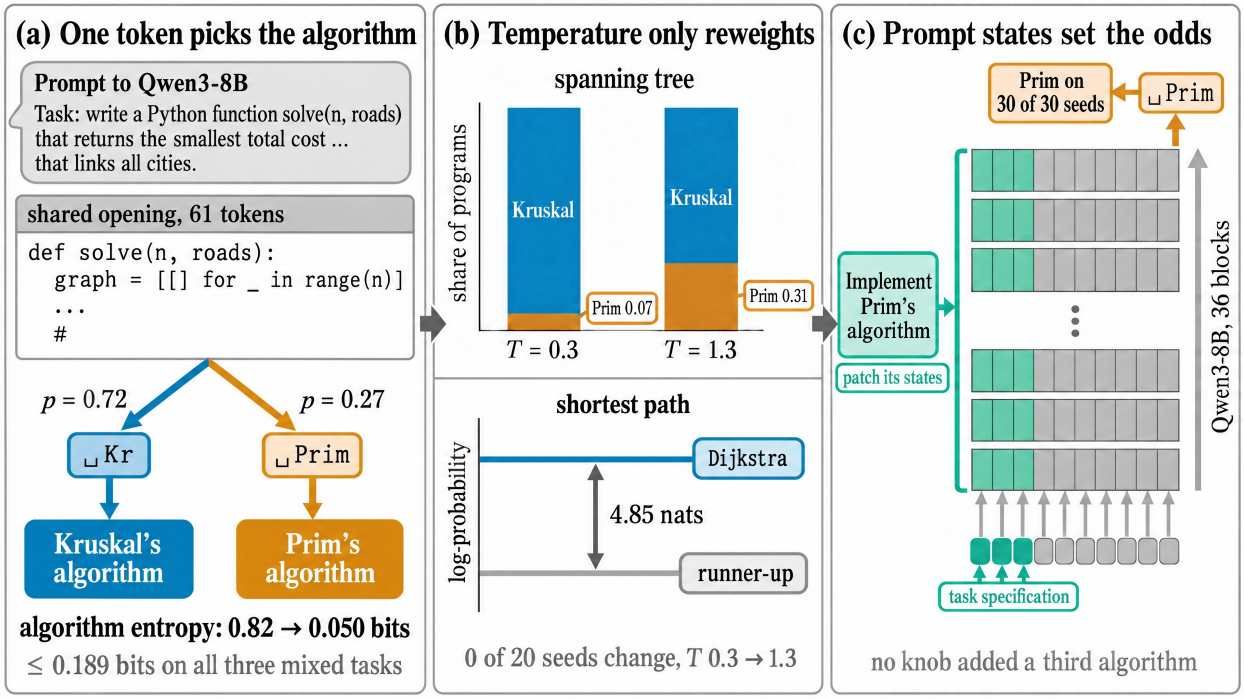


Figure 1: One forking token fixes the algorithm Qwen3-8B writes. (a) After a shared opening (grey), `␣Kr` leads to Kruskal (blue) and `␣Prim` to Prim (orange). (b) Temperature only reweights them; on shortest path, Dijkstra's 4.85-nat lead holds all 20 seeds. (c) Patching in task specification states (green) gives Prim on 30 of 30 seeds. Drawn overview; numbers from Section 5.

tokens [Qi et al., 2025], aligned models are less diverse than base models [Kirk et al., 2024, West and Potts, 2025] and concentrate probability on fewer continuations [Yang et al., 2025b], and in code, self-distillation separates forks that need exploration from locks that need precision [Zhang et al., 2026]. We add which algorithm follows forks in code, and the post-training margin there.

Activation interventions and geometry. Steering vectors [Rimsky et al., 2024] and single directions for refusal [Arditi et al., 2024] or truth [Marks and Tegmark, 2024] edit the residual stream, activation noise and steering raise output diversity [Khalid et al., 2026, Zhu et al., 2026], and prompt states encode attributes of the coming response [Dong et al., 2025]. Distributed alignment search (DAS) fits causal subspaces of low rank [Geiger et al., 2024, Wu et al., 2023], but subspace patching can act through pathways the model does not use [Makelov et al., 2024], and expressive maps can align a network with algorithms it does not implement [Sutter et al., 2025]. Sparse autoencoders [Lieberum et al., 2024] often steer worse than simple baselines [Wu et al., 2025b]; manifold steering follows fitted manifolds [Wurgaft et al., 2026] measured by geodesic distances [Tenenbaum et al., 2000]. We ask which site sets the forking token's odds.

Program search with an evaluator. FunSearch [Romera-Paredes et al., 2023], Evolution of Heuristics [Liu et al., 2024], ReEvo [Ye et al., 2024] and AlphaEvolve [Novikov et al., 2025] pair a language-model proposer with an executable evaluator, and many bin packing heuristics found this way reduce to simple parametric rules [Herrmann and Pallez, 2026]. We ask only whether selection yields new heuristics from an 8B model.

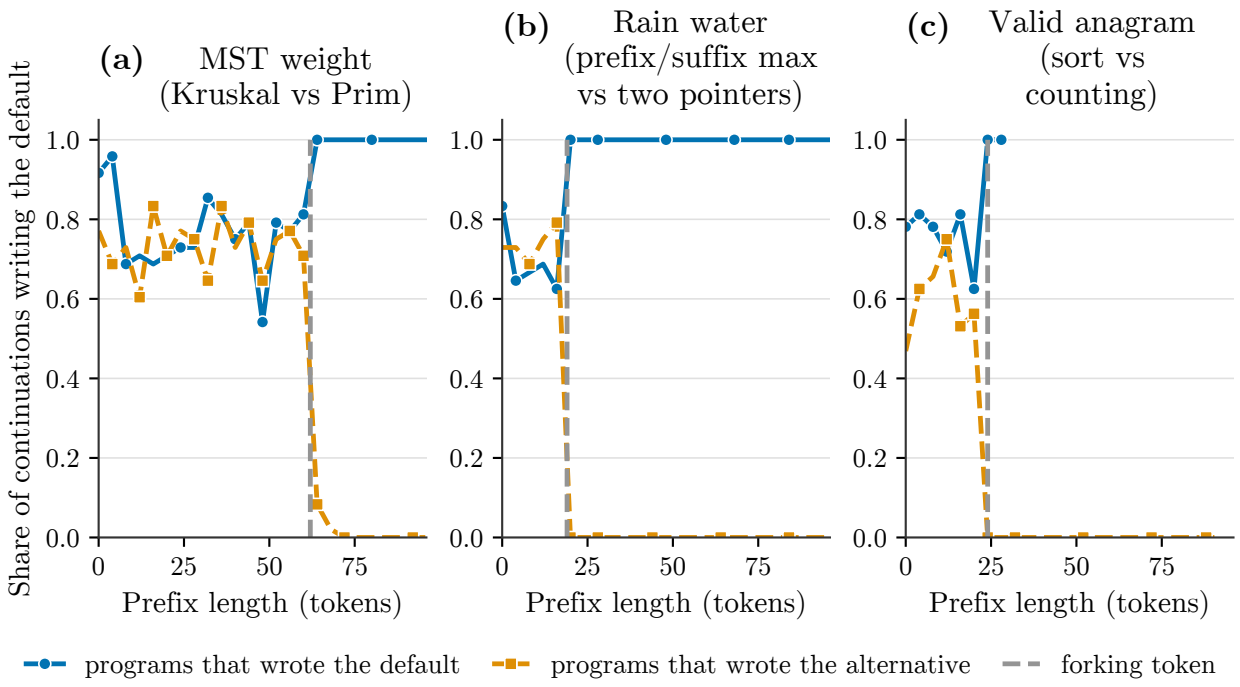


Figure 2: Prefix resampling locates a single forking token in Qwen3-8B (temperature 1) across three coding tasks. Each point is the mean, over 8 resampled continuations per program and 6 sampled programs per line (4 for valid anagram), of the share writing the task's default algorithm; both lines settle to 1 or 0 within a few tokens of the dashed forking token.

3 Method

Labelling programs. We call a task paired with a model a unit. Each task has a catalogue of three to ten known algorithms (Appendix A). A program is correct if it passes all of its task's hidden tests (40 to 48 per path task, 684 across the 13 bench tasks of Section 4) in a sandbox with a time limit per test case. A runtime tracer of operations on data structures, grid reads, edge relaxations and recursion assigns the mechanism, rules over the abstract syntax tree assign a second label, and a blind language-model judge sees only programs on which the two disagree. The combined label agreed with adjudicated gold labels at Cohen's kappa 0.966 on the path tasks (Appendix A). A program matching no catalogue entry is labelled other and audited for novelty (below). A unit's default is the algorithm its model writes most often; a unit is mixed when the model writes two algorithms, each in a substantial share of programs, and collapsed when it writes one in nearly all of them.

Locating the forking token. Following Forking Paths [Bigelow et al., 2025], prefix resampling keeps the first j generated tokens (the prefix length) of programs the model sampled itself, for $j = 0, 4, 8, \ldots$, draws 8 continuations at temperature 1 from each prefix, labels them, and tracks the outcome distribution over algorithms. A Bayesian change-point model with a permutation null marks where the share of the sampled program's own algorithm jumps by at least 0.4, and 20 further continuations there and at control positions confirm it, because estimates from few samples are noisy [Bigelow et al., 2026]. We call the confirmed position the fork and the tokens before it the opening. At the fork we read the teacher-forced next-token distribution and group candidate tokens into classes by the algorithm that continuations forced to begin with them write. The entropy $H(A \mid C)$ of the algorithm A given the class C of the sampled forking token measures how

much of the choice the token settles.

Predicting temperature's effect. The forking-token prediction at temperature T composes the tempered probability of staying on the opening, the tempered distribution over forking tokens, and the algorithm each forking token leads to. It was committed by its hash before any sample at the target temperatures and is scored by the total variation distance between predicted and observed algorithm distributions, averaged over $T \in \{0.3, 0.7, 1.0, 1.3\}$, against a baseline that predicts no change (the $T = 1$ shares at every T), tempering of those shares themselves, and a variant placed at the position of highest entropy. On collapsed units we read the forking-token margin, the gap in log probability, in nats, between the token class of the default and the strongest alternative class.

Comparisons under shared seeds. Paired comparisons use common random numbers: each sample slot, which we call a seed, owns one stream of uniform random numbers, and the sampler maps the tempered next-token distribution to a token by inverse-CDF lookup, so two programs from one seed differ only by the condition. We count the seeds whose algorithm differs across a set of conditions, compare paired outcomes by exact McNemar tests on discordant seeds [McNemar, 1947], and test contrasts over seeds by sign flips.

Perturbing and patching task specification states. A prompt has a condition sentence (empty, neutral or a ban, a negative constraint naming algorithms not to use), a task specification (problem statement and function signature) and a read position, its last token; we call the residual stream states at the tokens of the task specification its task specification states. Each ban has a neutral twin of identical token length in which irrelevant nouns replace the algorithm names. We call the following intervention a random push: it writes a random unit direction u into the residual stream during the prompt pass only. At block l and position i of the task specification, the component along u, measured from the neutral twin's state, is set to $\alpha\, n_{l,i}$, where $n_{l,i}$ is the norm of the displacement that the ban sentence itself produces there. One ban unit ($\alpha = 1$) is thus a random displacement of the size that a ban of one sentence causes; unless stated, it is applied at blocks 12 to 23. Activation patching replaces the task specification states of a prompt, at every block, with those of another prompt of the same length, for example `Implement Prim's algorithm` patched into a padding twin whose instruction tokens are filler; a random replacement of matched norm is the control.

Diversity and novelty. Diversity is the Hill number of correct programs, that is, the Hill number of order 1 [Hill, 1973], the exponential of the Shannon entropy, of the algorithm labels of correct programs, rarefied to subsamples of 25. Phrasings are clusters of programs whose AST 4-gram shingles, after identifiers are normalised, reach a Jaccard similarity of 0.8. A correct program labelled other counts as novel only if, after stress tests, a comparison with its nearest known solutions by tree edit distance [Zhang and Shasha, 1989], two blind judges gated on known control programs and a full manual read, it implements no known algorithm; with none among n programs, the rule of three bounds the rate by $3/n$.

4 Experimental Setup

Models. The main model is Qwen3-8B [Yang et al., 2025a] in non-thinking mode at temperature 1, in bfloat16 with exact common random numbers, and in some experiments in a W8A8 engine, with 8-bit floating-point weights and activations, on smaller GPUs; the perturbation results hold in bfloat16 (Appendix B). Qwen3-8B-Base serves for the base comparison, and gemma-3-4b-it, gemma-3-4b-pt [Gemma Team et al., 2025] and Llama-3.1-8B-Instruct [Grattafiori et al., 2024] as further families. The ban screen adds Qwen3-14B and Qwen3-32B in thinking and non-thinking mode, gemma-3-27b-it and Qwen2.5-Coder-32B-Instruct (`https://huggingface.co/Qwen/Qwen2.`

5-Coder-32B-Instruct).

Tasks. The two path search tasks follow a public set of creativity probes for code models (https://github.com/DrejcPesjak/llm-creativity): grid path search with moves to the four neighbouring cells, where any valid path is accepted so BFS is not forced, and weighted shortest path between one pair of nodes on a directed graph. Thirteen bench tasks derived from LeetCode problems [Xia et al., 2025], such as k-th smallest element, two sum and connected components, carry 684 hidden test cases and references for 54 algorithms. Sixteen further units, among them minimum spanning tree weight, trapped rain water and valid anagram, were built the same way [code] [code]. Each task has five prompt templates. Appendix G describes the online bin packing and TSP tasks of the program search.

Preregistration and audits. Every decisive rule and prediction was committed by its hash, in a SHA-256 chain, before the samples it governs, every headline number was recomputed from raw rows by an independent script, and placebo versions of each test were checked to fail. Shares carry Clopper-Pearson [Clopper and Pearson, 1934] or Wilson [Wilson, 1927] intervals, contrasts bootstrap intervals. Each experiment fits on one GPU, such as an RTX 4090 or an RTX 2000 Ada, in about 1 to 3 hours of GPU time, with vLLM [Kwon et al., 2023] for unpaired sampling. Code, prompts, raw generations, labels and the scripts that recompute each number are in the paper's repository [code] and linked at each claim.

5 Results

5.1 One forking token fixes the algorithm

On the three mixed units of Qwen3-8B, sampled programs share an opening of 18 to 61 tokens, and the next token settles the algorithm (Figure 2, Table 1): the share of continuations writing the default stays between 0.47 and 0.96 along the opening and jumps to 1 or 0 at the forking token. The preregistered test required the entropy of the algorithm given the class of the forking token to be at most 0.3 bits on at least two units. It passed on all three: 0.050 bits (bootstrap 95% CI 0.014 to 0.096) on minimum spanning tree weight, 0.153 (0.091 to 0.203) on trapped rain water and 0.189 (0.123 to 0.239) on valid anagram, against 0.82, 0.97 and 1.02 bits at the fork before the token is sampled. On the least decisive unit the margin to the bar is -0.111 bits (95% CI -0.177 to -0.061) [code]. On minimum spanning tree weight the opening ends with a comment marker; the next token is ␣Kr with probability 0.72, after which every continuation writes Kruskal's algorithm, or ␣Prim with probability 0.27, after which 97% write Prim's (␣ marks a leading space).

A blind locator on fresh samples recovered the forks of minimum spanning tree weight and trapped rain water and, on valid anagram, the next decision inside the counting branch (at most 0.004 bits), and found single forking tokens on two held-out gemma-3-4b-it tasks (0.001 bits each) [code], and Gemma's grid task forks the same way (Table 1) [code]. Collapsed units have forks too, which surface in perturbed and banned samples (Appendix A).

The fork is not universal. On Llama-3.1-8B-Instruct's grid task the openings that reach a forking token cover about 2% of programs, and the algorithm keeps 0.63 bits given the located decisions; on weighted shortest path, perturbed generations stay undecided for at least 60 tokens, and on k-th smallest the algorithm is fixed before the first generated token. Where a fork exists it is a high-entropy position (Appendix A).

5.2 Temperature only reweights the forking token

If one token decides, temperature can act on the algorithm only through that token's distribution and the probability of reaching it. Figure 3 compares the forking-token prediction, committed before sampling, with the observed share of the minority algorithm at four temperatures on shared

Table 1: Forking tokens located by prefix resampling (Qwen: Qwen3-8B; Gemma: gemma-3-4b-it; Llama: Llama-3.1-8B-Instruct). H: the algorithm's entropy in bits at the fork, before the token is sampled and given the sampled token's class (preregistered bar 0.3); the blind relocation used fresh samples and allows a second decision within two tokens.

Task (model)	Opening (tokens)	Forking token (probability) → algorithm	H at fork	H given token [95% CI]	Blind relocation
Minimum spanning tree weight (Qwen)	61	`␣Kr` (0.72) → Kruskal; `␣Prim` (0.27) → Prim	0.82	0.050 [0.014, 0.096]	0.004
Trapped rain water (Qwen)	18	`␣n` → prefix/suffix maxima; `␣left` → two pointers	0.97	0.153 [0.091, 0.203]	below 0.001
Valid anagram (Qwen)	23	`␣return` → sort; `␣count` → counting	1.02	0.189 [0.123, 0.239]	below 0.001 (count branch)
Substring search (Gemma, held out)	31	`for` (0.73) → naive scan; `try` (0.10) → library call	–	–	0.001
First and last position (Gemma, held out)	12	`first` (0.95) → linear scan; `left` (0.05) → binary search	–	–	0.001
Grid path search (Gemma)	32	`␣visited` → DFS; blank line, `if`, `queue` → BFS	–	–	–
Grid path search (Llama, held out)	52	`␣visited` (0.45) → DFS; `␣queue` (0.25) → BFS	–	–	0.629

token streams. Lowering the temperature from 1.0 to 0.3 removed the minority algorithm in paired seeds as predicted: all 18 discordant seeds moved to the default on minimum spanning tree weight (exact McNemar $p = 3.8 \times 10^{-6}$), as did all 10 on Gemma's substring search and all 13 on its first and last position.

The prediction's mean total variation distance from the observed shares (F in Figure 3) was 0.014 to 0.054 on five of six units, under the preregistered bar of 0.10, and 0.142 on Llama's grid task, whose many openings it does not model. Predicting no change (B) was almost as good: the forking-token prediction beat it on only 4 of 6 units (exact sign test $p = 0.34$), though it beat tempering of the shares themselves and the variant at the position of highest entropy on 5 of 6. A simpler prediction from the forking token alone, committed before sampling on the three mixed units, matched the observed response only on minimum spanning tree weight, one unit against the two its test required. The forking token explains the direction and mechanism of temperature's effect, but not yet its size.

On collapsed units the forking-token margin puts the alternative out of reach. On the neutral twin prompt the default leads the strongest alternative by 7.46 nats on grid path search, 4.85 on shortest path, 17.04 on connected components and 6.10 on k-th smallest. At $T = 1.3$ these margins imply rates of leaving the default of 0.002, 0.020, below 0.0001 and 0.009, and 100 continuations from each fork at $T = 1.3$ gave none. Table 2 tests the consequence with a preregistered design on shared seeds [code].

No temperature from 0.3 to 1.3 changed the algorithm of any seed on any task (upper 95% confidence bound on the rate per seed 0.17), in thinking mode either [code], although phrasings per algorithm grew with T (pooled slope 0.64 per unit temperature in log phrasings at a Jaccard threshold of 0.7, 95% CI 0.30 to 0.89). Under the same seeds, perturbations of one ban unit changed the algorithm, somewhere among the unperturbed run and three perturbed runs, for 17.25 of 20 seeds on shortest path (to Bellman-Ford or SPFA) and 4.55 on k-th smallest (to quickselect): a pooled contrast with temperature over seeds of 0.545 (95% CI 0.41 to 0.67; $p = 1.0 \times 10^{-5}$ by sign flips). The perturbation's own preregistered bar of 5 seeds per task failed on k-th smallest, whose

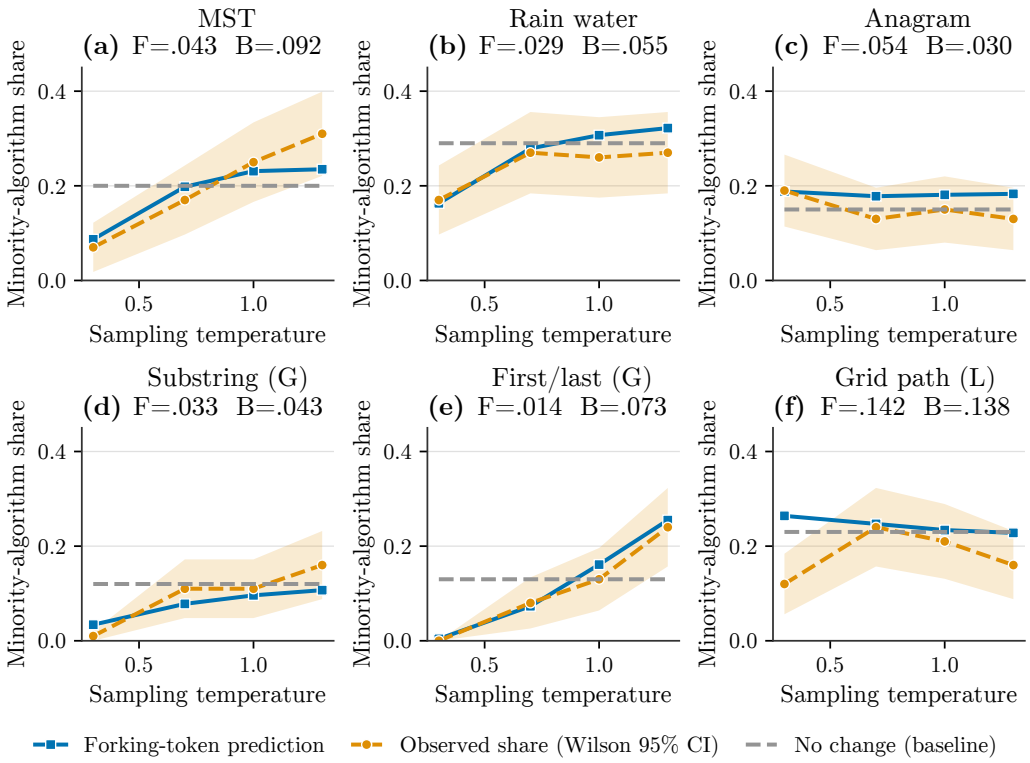


Figure 3: Lowering the sampling temperature shrinks the share of the minority algorithm roughly as the forking-token prediction implies, but the flat baseline of no change is often nearly as close to the observed points, and on the held-out Llama panel (f) the observed share diverges from both. F, B: each curve's mean total variation distance from the data. G: gemma-3-4b-it; L: Llama-3.1-8B-Instruct; others Qwen3-8B; $N = 100$ per cell.

Table 2: Seeds (of 20) whose algorithm changes across each row's conditions under common random numbers (Qwen3-8B, non-thinking unless stated; temperatures 0.3, 0.7, 1.0 and 1.3). Random push counts: seeds whose algorithm differs anywhere among the unperturbed run and three perturbed runs, averaged over the 20 subsets of three among six directions. Last row: contrast with temperature over seeds, bootstrap 95% interval.

Conditions compared	Shortest path (Dijkstra)	k-th smallest (full sort)	Grid path (BFS)
Temperature 0.3 to 1.3 (top-p 1)	0	0	0
… plus top-k 10 at T 0.7	0	0	0
… plus min-p 0.1 at T 2.0	1	1	0
Thinking mode, T 0.3 to 1.3	0	–	0 of 10
Random push, 1 ban unit	17.25	4.55	2.5
Random push, 2 ban units	19.05	10.6	8.7
Contrast at 1 ban unit [95% CI]	0.863 [0.775, 0.943]	0.228 [0.090, 0.380]	0.125 [0.025, 0.225]

dose unit was 0.77 times that of the screening experiment that selected the task (10.6 seeds at two ban units); on shortest path it cost correctness (pass rate 0.63 against 1.00).

At a fixed seed, then, tight temperature with top-k 10 yields about one idea in a few phrasings; top-k 10 even gave programs identical token for token to top-p 1 in 600 of 600 pairs that share a seed, since every forking token lies in the top ten. At $T = 0.3$, ten trajectories on minimum spanning tree weight were nine Kruskal programs in three phrasing clusters and one Prim program. Loosening the temperature moves mass to the second idea the forking token already weighs and multiplies phrasings (Kruskal from 4 to 22 clusters per 100-program cell between $T = 0.3$ and 1.3, Prim from 2 to 7); no third algorithm appeared.

5.3 The task specification states set the forking token's odds

Two measurements locate what sets the forking token's distribution. The first is activation patching. Patching the task specification states of a prompt that names the minority algorithm into its padding twin wrote the minority on 30 of 30 seeds on all three mixed units, every patched program correct, against 7, 4 and 0 of 30 for the padding twin and 12, 4 and 0 for a random replacement of matched norm. It moved the teacher-forced forking-token margin by -4.9, -5.5 and -22.1 nats, against -1.2, -2.0 and $+0.3$ for the replacement [code]. Patching the states of a ban of Dijkstra's algorithm into its neutral twin moved 24 of 30 seeds to Bellman-Ford or SPFA (21 correct; McNemar 24:0), whereas patching the read position transferred nothing on any of six tasks [code]. On two sum and find the duplicate the ban is carried instead by the condition sentence: patching its states transferred the ban on 30 and 27 of 30 seeds, patching the task specification states on 13 and 1.

The second is the random push, whose effect depends on its site. Figure 4 counts the correct programs off the default after the same random push, at the same dose per position, is written into six sites [code]. At the read position and on the condition sentence it changed the algorithm of none of 526 and 527 correct programs, and at the task specification's late blocks (24 to 35) of only 6 of 521. Yet a push at the read position displaces the final state as much as a push at the task specification (median norm 141 against 159), and at matched displacement the task specification keeps an advantage in log odds of 2.15 (95% CI 1.79 to 2.51). Pushes on the first 32 generated tokens, where the forks lie, also worked. The effects reappear in bfloat16 (Appendix B).

The perturbation reaches only known alternatives, and only on some tasks. With 40 fresh directions per task (8 on nine bench tasks), two ban units changed the algorithm on k-th smallest (63% of directions; quickselect or a size-k heap), connected components (73%; DFS or BFS), shortest path (20%; Bellman-Ford or SPFA), two sum (13%; brute force) and grid path (13%; DFS), and on none of the other ten tasks [code]. No subspace we fitted predicts which directions work (Appendix B), and cheaper knobs that act on the text reach as high a Hill number of correct programs (Appendix C).

5.4 Post-training widened the margin at the forking token

Is the default the base model's average answer? We scored identical token ids, formatted with the chat template, under Qwen3-8B and Qwen3-8B-Base (Appendix D). At the forking tokens of the three collapsed units the instruct margin exceeds the base margin by 3.27 nats on grid path search, 13.41 on connected components and 2.11 on shortest path, while the entropy at phrasing positions, which change only the wording, differs by only 0.23 to 0.31 nats, as preregistered [code]. The default is a sharpened choice, not the base model's mean: the base model already prefers it at the fork, by 2.55 to 3.50 nats. On minimum spanning tree weight post-training even reversed the preference: pushed through the instruct model's algorithm for each token, the base model's forking-token distribution implies Prim with probability 0.68, where the instruct model writes Kruskal in 78% of programs; Gemma's grid task shows a similar reversal [code]. Part of this concentration

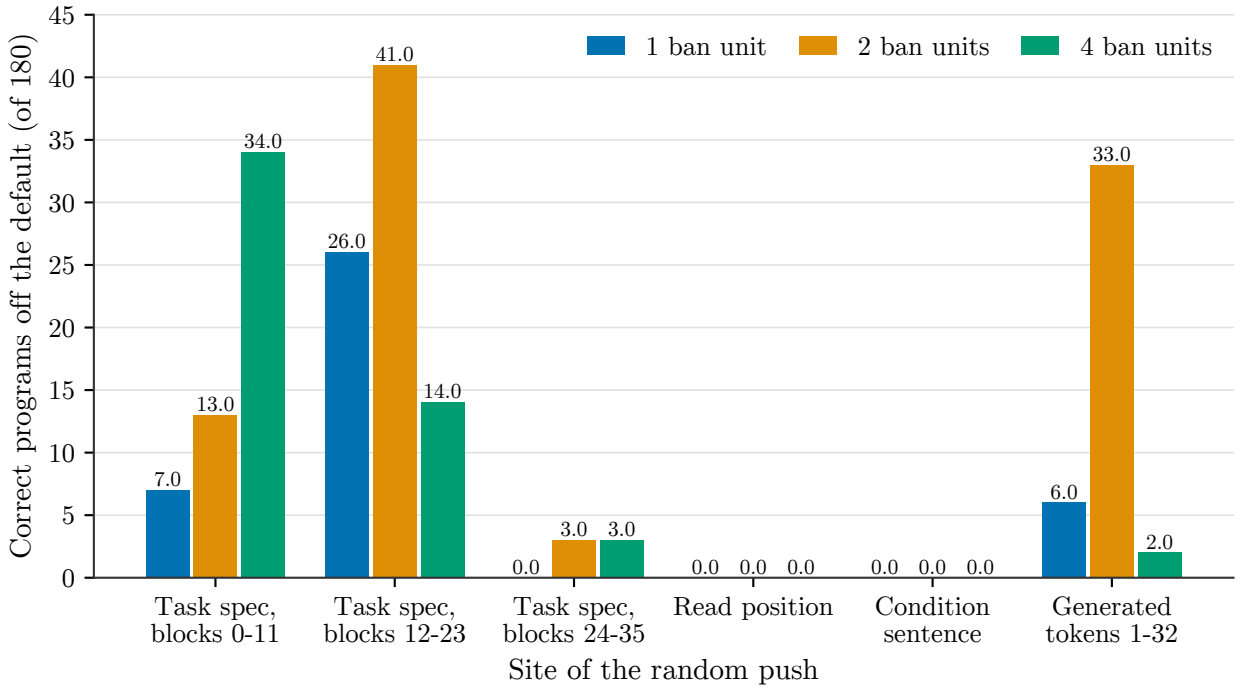


Figure 4: On Qwen3-8B (W8A8 engine), a random push changes which algorithm gets written mainly in the task specification states of blocks 0–23 or the first 32 generated tokens; pushes at the read position or condition sentence never do. Bars count correct programs off the default, pooled over six coding tasks (180 samples per bar); the drop at 4 ban units reflects broken programs, not reversion to the default.

reflects base-model uncertainty at the fork (Appendix D) [code].

5.5 Prompts move the choice among known algorithms

Context seeds and instructions. Prepending a context seed, random words (`seed=(horse chair roof love)`) or a random string (`seed=d32qxk4nt43`), shifted the modal algorithm by at most one percentage point on both path tasks (90% intervals within ±15 points). The two long novelty instructions (`Make a fully novel algorithm never seen before`, with a 2,000-word rationale) made 84% of 574 outputs claim a new algorithm. Yet 39% of those contained no code, 75% of the rest implemented catalogue algorithms, and none of the 23 correct programs outside the catalogue was new. The instructions spread the choice over known algorithms (Hill numbers of correct programs 1.8 to 4.5 without thinking) at a cost in pass rate [code]. On bin packing, `Make a new algorithm` returned best fit in 28 of 29 and 28 of 28 programs of Qwen3-14B and Qwen3-32B in thinking mode [code].

Bans. A ban of the default is mostly obeyed and lands on one known substitute: DFS on the grid task (86 of 100) and Bellman-Ford or SPFA on shortest path (83 of 100) [code]; on the bench tasks it acts where the forking tokens surface (Appendix C). A ban of the five grid algorithms used most is answered differently: Qwen3-8B avoided none of them and reimplemented them without their names in 41% of programs. Across ten configurations of open-weight models, lexical compliance (no banned name in the program) exceeded mechanistic compliance (a search mechanism outside the banned set) by 0.70 (95% CI 0.57 to 0.82), positive in all ten (descriptive; a second instrument missed its agreement threshold) [code]. Only Qwen3-32B in thinking mode obeyed often enough to clear the preregistered bar of 0.20 (17 of 50 correct survivors, programs that pass and leave the banned set; Table 5).

Edge or opposite. Figure 5 places Qwen3-32B's baseline and ban prompts among the task specification states of forced prompts for ten grid algorithms. Along the first principal component the ban moves the prompt states from the BFS and DFS end toward the middle of the banned set, not toward Dijkstra with unit weights, which we name UW-Dijkstra as in the figure, although every catalogue survivor of the obeying model is Dijkstra with unit weights (6 of 6; 9 of 9 on held-out templates). Whether that survivor lies at the edge of the banned five or opposite them depends on the map. On an Isomap graph [Tenenbaum et al., 2000], classical multidimensional scaling of geodesic distances, its geodesic distance from the banned five is 1.63 times the median of the algorithms outside the banned set on a manifold built from prompts that name the algorithm (opposite) but 0.67 on one built from programs with the names removed (edge; Appendix E) [code]. The geometry is correlational: pushing prompt states toward the survivors' centroids produced BFS in all 300 programs, and no manifold showed a ring.

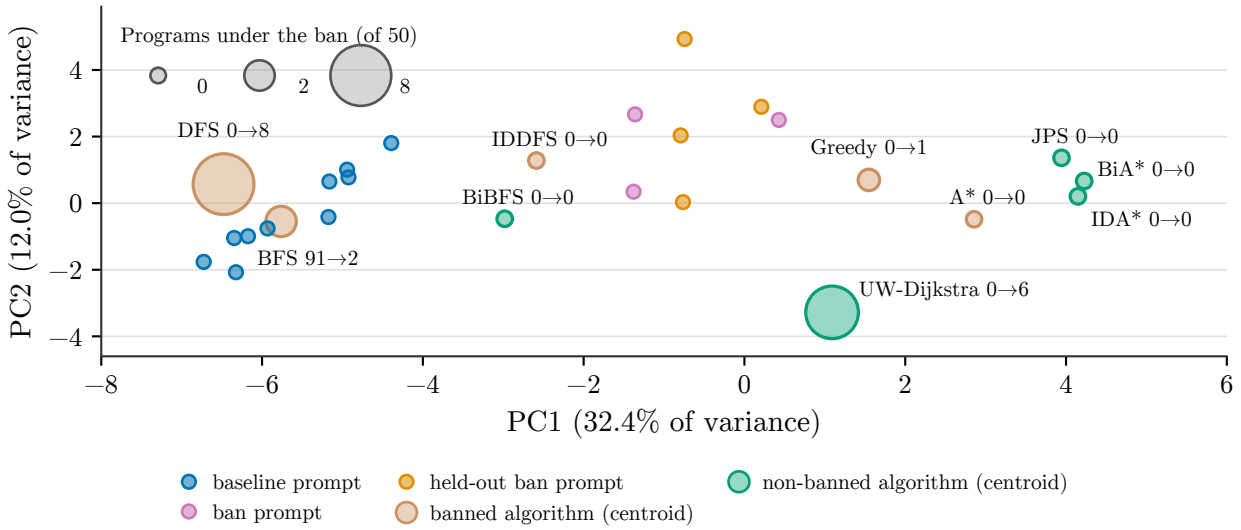


Figure 5: Qwen3-32B (thinking) task specification states at block 31, first two principal components. Labelled circles: centroids of 300 forced prompts, sized by programs written under the ban of five names (labels give baseline → ban counts). Small circles: baseline (10), ban (3) and held-out (4) prompt states. Baseline states sit at BFS and DFS; Dijkstra with unit weights, the only catalogue survivor of the ban, sits apart.

5.6 No route produced a confirmed new algorithm

In the nine experiments whose audits cover all of their correct programs, none of 45,892 correct programs implemented a new algorithm (bound by the rule of three 0.007% per program); the 39 correct programs labelled other were labeller misses, catalogue variants or degenerate (Table 8 in Appendix F). Forcing each of the forking token's top 64 candidates reached only one more known algorithm on each of three units, about 2 to 50 million samples away at $T = 1$. An evaluator helps only in part: with selection, an island-model search with Qwen3-8B as mutator beat best fit on held-out bin packing instances (1.40% to 2.42% excess bins on a Weibull test set, against 3.98%), and an independent implementation reproduced the gain for three of five arms [code]. None beat the strongest published heuristics. One reader classed eight programs of the first search, all preferring a leftover that is a whole multiple of the item size, as a new idea, but only one of three judge families passed its control gate, short of the preregistered two, and the replication found none of its 40 audited candidates new (Appendix G).

6 Discussion and Limitations

One forking token accounts for these results. Decoding knobs act on its distribution and cannot cross a wide margin; prompts that name an alternative, and on the path tasks bans, act on the task specification states that set its odds; and post-training wrote the default as a wide margin at this token. Read this way, a code model's creativity in choosing an algorithm is the entropy of one token over a few known options, and every knob we tested only moves mass among them. The data reject the view that the model returns the mean of its possible answers and that only context moves it elsewhere: the default is a sharpened, sometimes reversed, base-model choice, and context moved the algorithm only by changing the states that set the token's odds. Consistent with Lee et al. [2025], temperature above 1 raised the Hill number of correct programs modestly on mixed units, not on collapsed ones (Appendix C).

For practice, this suggests acting on the forking token's inputs rather than on global temperature; for evaluation, it argues for labels assigned by execution and for designs with shared seeds, without which rewritten code looks like a new idea. Internal readouts locate the decision but certify no variable for it (Appendix H), as Makelov et al. [2024] and Sutter et al. [2025] warn.

6.1 Limitations

The main model is Qwen3-8B in non-thinking mode. Forking tokens were confirmed in gemma-3-4b-it but not in Llama-3.1-8B-Instruct, whose grid choice is reached through many openings, and the account does not cover every task: shortest path decides late and k-th smallest before generation. Forks can be located by resampling only where the model already writes two algorithms; on collapsed units they were found in perturbed or banned samples, with margins from teacher forcing. The forking-token prediction of temperature's effect is not established as better than predicting no change, and only three held-out units were scored, one fewer than preregistered.

The preregistered contrast with shared seeds ran in a W8A8 engine on two tasks that an earlier screening experiment had selected because random pushes moved them; it was not rerun in bfloat16, although effects under 40 other directions reproduced there, and the perturbation moves the algorithm on only 5 of 15 tasks. The tasks are well-known problems with small catalogues of textbook algorithms the models have likely memorised, so open-ended design problems may behave differently. Two instruments missed their preset agreement thresholds, so the results for context seeds and the compliance values are descriptive. The novelty audits rest on catalogue labels for most programs and on one reader's manual reads for disputed ones, judge agreement was often low, and the program searches used at most 2,000 calls per arm and seed, far fewer than published discovery systems. The internal handles are not specific variables, and the geometry of Section 5.5 is correlational.

7 Conclusion

Where it could be located, the algorithm an open-weight code model writes was chosen at one sampled token, among options the model already holds. That explains why temperature rephrases code without changing its algorithm on collapsed units, while prompt cues and edits of the task specification states can change it. Making a model creative in this sense means placing a new option at that token, which no knob we tested did. Future work should test the account on more model families and on tasks without a textbook catalogue, predict the size of temperature's effect from openings summed over many paths, and give program-search operators that act on the forking token's inputs.

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A Tasks, catalogues and labeller validation

Table 3 lists the tasks discussed in the main text with the default each model writes and the catalogue against which programs are labelled. Prompts state the task, the function signature and a format instruction of one line; five templates per task vary the wording. The grid task accepts any valid path, so BFS is not required for correctness.

Table 3: Tasks, defaults and catalogues (Qwen3-8B unless stated).

Task	Default	Catalogue
Grid path search	BFS (gemma-3-4b-it: DFS 70%)	BFS, DFS, iterative deepening DFS, A*, greedy best first search, bidirectional BFS, Dijkstra with unit weights, IDA*, jump point search, bidirectional A*
Weighted shortest path	Dijkstra	Dijkstra, Bellman-Ford or SPFA, bidirectional Dijkstra, Floyd-Warshall, Dial
k-th smallest element	full sort	full sort, size-k heap, heapify and pop, quickselect, median of medians, counting sort
Substring search	naive scan (gemma-3-4b-it also)	naive scan, KMP, Rabin–Karp, Z algorithm, Boyer–Moore–Horspool, library find
Two sum	hash map	hash map, sort and two pointers, brute force, sort and binary search
Find the duplicate	hash set	hash set, sort and compare neighbours, Floyd cycle detection, binary search on values, index marking
Connected components	union–find	BFS, DFS, union–find, label propagation
First and last position	binary search (gemma-3-4b-it: linear scan)	binary search, bisect module, exponential search (gemma-3-4b-it adds linear scan)
Minimum spanning tree weight	Kruskal (0.80)	Kruskal, Prim with a heap, Prim with an array scan, Borůvka
Trapped rain water	prefix and suffix maxima (0.68)	two pointers, prefix and suffix maxima, monotonic stack, split at the highest bar
Valid anagram	sort and compare (0.77)	sort and compare, hash map of counts, fixed 26-slot count array

Labeller validation. Every labeller was checked before use on reference implementations of every catalogue algorithm and on mutated programs that must fail the hidden tests. On the path tasks the runtime tracer and the syntax rules each labelled 22 of 22 references correctly; on evaluation gold sets runtime and syntax labels agreed at kappa 0.753, the runtime label agreed with adjudication at 0.843 and the combined label at 0.966 (the runtime label missed the accuracy targets of that first gate, so every disagreement between the two labels went to the blind judge). On the 13 bench tasks all 54 references passed at no more than a quarter of their time limits, 39 mutants failed, runtime

and syntax labels agreed at kappa 0.84 to 1.00, and the runtime label agreed with adjudication at 0.91 or more. Repeated style checks, in which two blind judges label 30 programs per unit, agreed with the frozen labellers on 30 of 30 programs for each of the three mixed units. Two instruments did not pass their gates and their results are reported as descriptive: the labeller of the screening experiment on context seeds and temperature (gold agreement 93% and 70% on the two path tasks) and the expansion-order tracer used for bans, which reads no names (182 of 260 gold programs, against a bar of 90%).

Forking tokens. On Qwen3-8B's connected components, a collapsed unit, the fork appears in perturbed and banned samples: after `␣def` the token `␣dfs` writes DFS and `␣find` union–find, and forcing `␣def␣dfs` after the neutral twin prompt gives correct DFS on 30 of 30 seeds [code]. In the forking-token experiments, resampling with no prefix kept reproduced each mixed unit's baseline minority share (Fisher p 0.19 to 0.58). A second, blind locator repeats the forking-token procedure on fresh samples and allows a second decision within the next two tokens; on Gemma's grid task, BFS follows `␣visited` in 1.1% of 1,409 continuations and the other forking tokens in 95% to 100%. A fork lies above the 90th percentile of per-token entropy in all 209 samples that reach it, though it is not always the single position of highest entropy [code] [code]. On trapped rain water `␣n` leads to prefix and suffix maximum arrays and `␣left` to two pointers; on valid anagram `␣return` leads to sorting and `␣count` to counting (Table 1).

B Further perturbation and patching results

Engine precision. When 40 fixed directions were re-run at the same absolute dose in bfloat16, connected components changed algorithm under 26 of them, against 29 in the W8A8 engine, and shortest path under 6 against 8, with Spearman correlations across directions of 0.96 and 0.87 and the same substitute algorithms [code]. A bfloat16 rerun of the site comparison of Figure 4 found the same order on k-th smallest (24:5 paired changes; Mantel-Haenszel odds ratio at matched displacement 4.6, 95% CI 1.9 to 11.3), although its preregistered site rule was withdrawn because the arm that pushes the condition sentence lacked common support.

Dose unit and site. The dose must be counted in the absolute ban unit of Section 3. A unit set as a fixed fraction of the neutral twin's residual norm moved connected components to a correct alternative in 1 of 50 seeds where the absolute unit did so in 25 of 50, and in a crossed design the unit accounted for a risk difference of 0.45 (95% CI 0.35 to 0.55) against 0.03 for the direction seeds [code]. Early blocks (0 to 11) moved 34 programs at four ban units in Figure 4, but their unit is a smaller fraction of the residual norm (4.5% to 7.2% against 12.7% to 18.3% for blocks 12 to 23), so early and middle blocks are not separated.

Heterogeneity and specificity. Single directions changed 3 to 17 of 20 seeds on shortest path and 0 to 7 on k-th smallest in the preregistered contrast of Table 2. Which directions work is reproducible across templates but not predicted by their projection on a subspace fitted to prompts that force an algorithm (pooled Spearman 0.03), and clamping a DAS subspace fitted at these states removed none of 218 algorithm changes [code].

Patching on shortest path. The patch from `Implement Bellman-Ford or SPFA for this task` into its padding twin gave Bellman-Ford or SPFA on 30 of 30 seeds.

C Further knob and ban results

An intent sentence written by Qwen3-8B-Base for the instruct model to implement reached a higher Hill number of correct programs than the random push on three of five tasks (Table 4), against the preregistered prediction that the random push would match the best plan sampler that works in

Table 4: Highest Hill number of correct programs reachable within 5 points of pass rate, by knob and task (Qwen3-8B, bfloat16, common random numbers, $N = 50$ per cell). The random push row uses a unit relative to the residual norm, which is weaker than the absolute ban unit; at the absolute unit it reaches 3.22, 2.13, 1.21, 1.10 and 1.39.

Knob	k-th smallest	Connected components	Shortest path	Two sum	Grid path
Random push	1.98	1.09	1.18	1.17	1.06
Intent sampled hot, code cool	1.19	1.96	1.00	1.00	1.00
Plan search before code	1.09	1.50	1.07	1.00	1.00
Enumerate algorithms, pick one at random	1.83	1.88	1.24	1.27	1.08
Base-model intent sentence	1.73	2.53	1.45	1.61	1.04
Temperature on tokens 1 to 32, matched in KL	1.10	1.25	1.09	1.00	1.00

text. Temperature 1.3 and top-k 10 reached at most 1.09 on every task, whereas temperature above 1 raised the Hill number of correct programs on minimum spanning tree weight, a mixed unit, from 1.82 to 2.07. In Figure 3 the minority share barely grows above $T = 1$ on the Qwen3-8B units, because samples that leave the opening mostly return to the default. At $\alpha = 2$ of the relative unit the perturbation moved the algorithm on 19 seeds that the temperature matched to its departure did not, and the reverse on 3 (exact $p = 0.0009$), 13 of the 19 on k-th smallest. Raising the temperature only at the forking token raised the Hill number of correct programs on valid anagram from 1.78 to 2.78 at a cost of 2 points of pass rate, but reached DFS on the collapsed connected components in only 3 to 9 of 50 programs. In thinking mode a repeat with the same seed differed at the byte level in all 20 shortest path traces yet kept every label, and adding top-k 10 changed none of 20 seeds. A short story implying an approach without naming it moved shortest path toward Bellman-Ford by 66 points in the screening experiment whose labeller missed its agreement threshold and in 15 of 50 programs in one bfloat16 run, but left Dijkstra in 30 of 30 programs in another.

Single bans. On the bench tasks a single ban is enforced where the forking tokens surface: forcing the default's opening up to its first body token undid the ban in 98% to 100% of samples, while a neutral function signature undid none [code]. Over the 12 bench tasks, the template of a single ban was the strongest predictor of whether it was obeyed (likelihood ratio statistic 374 on 4 degrees of freedom; a placebo with shuffled templates gave p 0.39), and the template `X is not allowed here.` failed on 10 of 12 tasks [code].

Bans of five names. Under the ban of five names Qwen3-8B wrote BFS in 63 and DFS in 36 of 100 programs. Among programs that used a banned mechanism, 55% to 100% contained no alias of any banned algorithm. Qwen3-32B with thinking survived through Dijkstra with unit weights (6), Bellman-Ford or SPFA run on the grid (7), frontier search with random priorities (2), a random walk (1) and wavefront propagation (1), and its mean reasoning length rose from 6,864 to 9,090 tokens under the ban; on held-out templates it kept 21 correct survivors of 50. The expansion-order tracer, which reads no names, puts the gap between lexical and mechanistic compliance at 0.61 (95% CI 0.45 to 0.76). With an exact behavioural instrument on bin packing, renaming appeared without thinking (Qwen3-8B, gap 0.40, 95% CI 0.21 to 0.55, but 0.08 on held-out templates) and not with it (+0.04 and −0.13 for Qwen3-14B and Qwen3-32B).

Bans under different instruments. Under cumulative bans of one to five grid algorithms with thinking on, the share of programs that left the banned mechanisms at five names depends on the instrument: for Qwen3-32B it is 0.22 by the runtime label, 0.59 by hand reads of 46 programs, 0.67

Table 5: The grid ban of five names across ten open-weight configurations. Each configuration's banned set was its own five algorithms used most, which was BFS, DFS, iterative deepening DFS, A* and greedy best first search for all ten. A correct survivor passes the hidden tests and uses a mechanism outside the banned set.

Configuration	Baseline programs	Correct survivors	Share [Wilson 95%]	Share with a banned mechanism
Qwen3-8B, thinking	BFS 30 of 30	0 of 30	0.00 [0.00, 0.11]	1.00
Qwen3-8B, non-thinking	BFS 100 of 100	0 of 100	0.00 [0.00, 0.04]	1.00
Llama-3.1-8B-Instruct	BFS 77, DFS 20, other 3	0 of 50	0.00 [0.00, 0.07]	0.90
Qwen3-14B, non-thinking	BFS 94, DFS 6	0 of 50	0.00 [0.00, 0.07]	1.00
Qwen3-14B, thinking	BFS 96, DFS 4	7 of 50	0.14 [0.07, 0.26]	0.76
Qwen3-32B, non-thinking	BFS 81, DFS 19	4 of 50	0.08 [0.03, 0.19]	0.82
Qwen3-32B, thinking	BFS 91, invalid 9	17 of 50	0.34 [0.22, 0.48]	0.22
gemma-3-27b-it	BFS 99 of 99	0 of 50	0.00 [0.00, 0.07]	1.00
Qwen2.5-Coder-32B-Instruct	DFS 52, BFS 48	0 of 50	0.00 [0.00, 0.07]	0.94
gemma-3-4b-it	DFS 70% (separate run)	0 of 50	0.00 [0.00, 0.07]	1.00

by the expansion-order tracer and 0.74 by its fine classes; Qwen3-14B's compliance gap exceeds 0.4 under every reading except the fine tracer on held-out templates [code]. Thinking traces name the banned rule in every bin packing cell, and Qwen3-32B with thinking moved to first fit in 14 of 30 programs under a best-fit ban.

D Base and instruct models at the forking token

Table 6 gives the margins behind Section 5.4, scored at the forks and at 64 phrasing positions of 20 programs per task. Its instruct margins come from 20 programs per task over five templates, so they differ from the margins of Section 5.2, which were read on the neutral twin prompt. The Jensen-Shannon divergence between the two models, that is, the mean KL divergence of each from their average, is 7 to 12 times larger at the forking token than at phrasing positions on the collapsed units.

Table 6: Forking-token margins (nats, default minus strongest alternative) of Qwen3-8B and Qwen3-8B-Base on the same token ids, the entropy change at phrasing positions, and the Jensen-Shannon divergence between the two models at the fork, at phrasing positions and at positions matched for the entropy of the base model.

Task	Instruct margin	Base margin	Gain	Phrasing entropy change [95% CI]	Divergence: fork / phrasing / matched
Grid path search	6.33	3.07	3.27	0.310 [0.292, 0.326]	0.507 / 0.053 / 0.274
Connected components	15.96	2.55	13.41	0.275 [0.269, 0.282]	0.306 / 0.042 / 0.203
Weighted shortest path	5.61	3.50	2.11	0.229 [0.223, 0.235]	0.440 / 0.037 / 0.286
Minimum spanning tree weight	1.04	−0.80	1.84	0.337 [0.326, 0.350]	0.342 / 0.051 / 0.340
Trapped rain water	1.50	−0.03	1.53	0.210 [0.200, 0.220]	0.111 / 0.028 / 0.201

On Gemma's grid task the base model puts 0.089 of the fork's mass on the DFS tokens against 0.671 for the instruct model, and the two disagree on the most likely forking token for 20 of 20 prompts [code]. The divergence at the fork is 0.55 to 1.85 times that at positions matched for base entropy in Qwen3-8B, and 0.75 to 1.18 times in Gemma. In the weights, restoring the base unembedding rows of the three forking-token classes of connected components closes 0.548 (95% CI 0.44 to 0.71) of the margin gap, against a median of 0.044 for isotropic edits of equal norm, almost all through the alternatives' rows (0.489) rather than the default's (0.060): post-training

suppressed the readout of `␣dfs` and `␣bfs` more than it boosted `␣find` [code]. Random edits of equal norm in the residual stream's top 64 principal directions close as much ($p = 0.37$), so the effect lies in the readout but is not specific to these tokens. In generation the edit gave correct DFS in only 5 of 30 seeds, short of the preregistered 10.

E Manifold maps of the ban of five names

Table 7 gives the position of the surviving algorithm, Dijkstra with unit weights, on four maps. The preregistered prediction of opposite on the two maps built from names held, and that of edge on both the program and the description maps failed on the description map [code]. Manifolds built from language track the lexical overlap of the names (Spearman -0.65 with geodesic distance) and the data structures that textbooks pair with each algorithm (representational similarity 0.65 against 0.46 for lexical distance on Qwen3-32B's named manifold), and the one built from programs encodes the data container more than the search discipline [code]. The centroid push that produced BFS in all 300 programs was applied to Qwen3-8B's prompt states.

Table 7: Geodesic distance (Isomap, $k = 15$ nearest neighbours) of the surviving algorithm, Dijkstra with unit weights, from the banned five, as a multiple of the median over the algorithms outside the banned set; above 1 reads as opposite, below 1 as edge.

Manifold built from the states of	Qwen3-32B	Qwen3-8B
Prompts naming the algorithm	1.63	1.17
Names minus a neutral twin matched in tokens	1.20	–
Descriptions without names, which the model names correctly	1.08	1.01
Correct programs with names removed	0.67	0.63

F Novelty audits

Table 8: Novelty audits that cover all correct programs of an experiment. A correct program labelled other is adjudicated by stress tests, nearest known solutions, gated blind judges and a manual read.

Experiment	Model	Correct programs	Correct, labelled other	New algorithms
Site comparison of the perturbation	Qwen3-8B	3,555	1	0
Forty fresh directions on 15 tasks	Qwen3-8B	10,651	4	0
New units, first set	Qwen3-8B and Base	6,854	2	0
New units, second set	Qwen3-8B and Base	10,366	14	0
Gemma forking token	gemma-3-4b-it	1,450	1	0
Temperature at the forking token	Qwen3-8B and Gemma	5,543	2	0
Distributed alignment search	Qwen3-8B and Gemma	1,857	0	0
Gemma subspace edit in free generation	gemma-3-4b-it	913	0	0
Temperature prediction	three families	4,703	15	0
Total		45,892	39	0

The experiment with the long novelty instructions audited 38 correct programs outside the catalogue, 23 of them written under those instructions: 30 were labeller misses (mostly counting sort with a dictionary), 7 catalogue variants and 1 degenerate, and none was new. None of 1,143 one-shot bin packing programs was both behaviourally novel and better than best fit. Forcing each of the forking token's top 64 candidates reached BFS on connected components, Prim with an

array scan on minimum spanning tree weight and an alphabet scan on valid anagram, all catalogue algorithms.

G Program search with an evaluator

The program search writes online bin packing heuristics as priority functions, the interface of FunSearch [Romera-Paredes et al., 2023], and constructive TSP heuristics scored by the optimality gap to LKH-3 tours [code] [code]. A bin packing heuristic is scored by its excess bins, that is, the percentage by which the bins it uses exceed the lower bound of total item size over bin capacity, rounded up, on two held-out sets: FunSearch's Weibull 5k test set and binpack3 of the OR-Library, the public collection of operations research benchmarks. Each island-model search used at most 2,000 calls per arm and seed.

Table 9: Best programs of the island-model search on held-out bin packing instances (excess bins, %, lower is better, on FunSearch's Weibull 5k test set and on binpack3). Arms: temperature 1.0; temperature 1.6 with top-p 0.95; a forced rank-2 to rank-8 token at the code position of highest entropy; the same forcing at a random position; an instruction to use a different idea.

Arm	Bin packing, seeds 0–1 (Weibull / binpack3)	Replication, seeds 2–3 (Weibull / binpack3)
Temperature 1.0	1.53 / 3.55	3.672 / 4.821
Temperature 1.6	1.40 / 3.55	equal to best fit
Forcing at the token of highest entropy	2.42 / 4.17	2.395 / 3.852
Forcing at random position	1.73 / 3.48	1.720 / 3.703
Instruction to use a different idea	1.85 / 3.38	2.133 / 3.554
No selection	3.98 / 5.37 (best fit)	–
Best fit; FunSearch heuristic	3.98 / 5.37; 0.68 on Weibull	same harness

Selection, not sampling, produced the gains in Table 9: the same calls without selection never improved on the seed program. Forcing a rank-2 to rank-8 token at the code position of highest entropy found more distinct behaviours, clusters of programs that agree on at least 95% of 2,000 probe decisions, than temperature 1.0 in both bin packing searches (3.0 and 17.8 times as many per call), but its preregistered advantage did not hold: it reached 1.2 times temperature 1.6 in the first search against a bar of 1.5, 0.74 times forcing at a random position in the replication, and 0.96 times temperature 1.0 on TSP. One reader classed eight bin packing programs of the first search as a new idea, but only one of three judge families classified its control programs well enough to count there, and the replication found none of 40 audited candidates new. On TSP every core arm's best closed at least 20% of nearest neighbour's 23.94% optimality gap on 200-city instances (bests 13.98% to 16.68%, closing 30% to 42% of the gap), but none beat the ReEvo/EoH seed rule (13.67%). A related search on k-th smallest found known techniques outside the catalogue (binary search over the value range, order statistics on a Fenwick tree, bucket select) at 7.8 times the rate per call of the instruction alone, short of a preregistered 10 times, and none new [code] [code].

H Internal readouts at the forking token

We read the decision with sparse autoencoders, a natural language autoencoder (a verbaliser that explains an activation in text and a reconstructor that maps the text back), distributed alignment search, circuits and lenses. Table 10 summarises what each found; none certified a specific variable for the algorithm, and sparse autoencoder features read Gemma's fork less well than the raw residual stream.

Table 10: Internal readouts of the algorithm choice.

Method	Model and site	Result
Sparse autoencoder readout (Gemma Scope 2, 16k latents)	gemma-3-4b-it, layer 29, fork	Lasso on latent codes predicts the log odds at the fork over 120 prompt variants at R^2 0.66 against 0.18 for text; ridge on the raw residual stream reaches 0.72
Sparse autoencoder steering	gemma-3-4b-it, layer 22, fork	logit effect at matched norm: difference in means +4.04 nats, feature clamp +1.78, random features +0.51, isotropic +0.12; edits toward BFS lower the pass rate from 0.84 to between 0.16 and 0.58, except a clamp of the top five features (18 BFS of 50, pass 0.84)
Sparse autoencoder transfer	Qwen-Scope base-model SAEs on Qwen3-8B	loss ratio with the reconstruction spliced in, 1.59 to 3.29 at layers 15, 20 and 27 fails the 1.5 transfer bar [code]
Natural language autoencoder	Qwen2.5-7B-Instruct, layer 20	at the last positions of the prompt the explanation never names the planned algorithm (0 of 112, as text alone); at the token of the algorithm name it names it in 80% to 89% [code]
Distributed alignment search, rank 4	Qwen3-8B connected components, token 11	edits toward DFS write correct DFS in 29 of 30; edits toward BFS write BFS in 29 of 30 but only 5 pass; maps fitted to shuffled labels keep 0.63 to 0.87 of the effect: no handle
Distributed alignment search, rank 4	gemma-3-4b-it grid, fork	correct BFS in 22 of 30 against 6 for the neutral twin and 5 for random subspaces: a sufficient handle that fails specificity controls
Distributed alignment search in free generation	gemma-3-4b-it grid	38 of 50 correct BFS against 13 of 50 (26:1 discordant pairs); free prompting gives 32 of 50 ($p = 0.21$) and the best wording 49 of 50 [code]
Path-patching circuit and cross-model patching	Qwen3-8B, fork	late MLPs and heads at the fork carry the forking-token logit; patching base-model activations into 10 validated nodes closes 1.00 of the gap between base and instruct margins on minimum spanning tree weight, against 0.81 for matched random late sets ($p = 0.45$), and 0.45 against -0.22 on connected components ($p = 0.020$)
Logit lens	Qwen3-8B, ban prompts	no layer vetoes the default at the program's signature token; the choice precedes it [code]
Jacobian lens	Qwen3-8B, layer 27	fails its gate against the logit lens (top-1 agreement 0.104 against 0.291)
Thought anchors	Qwen3-8B, thinking on	on the grid task under a ban the algorithm is fixed at the first resampled sentence in 4 of 4 traces